Using Frequency-Resolved Optical Gating (FROG) to Measure Ultrafast Laser Pulses

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It's hard to measure the spectral phase of an ultrashort laser pulse because we need a shorter event as a reference to measure an event in time. So the problem is, how do we measure the shortest event in time? In our case, how do we measure ultrafast laser pulses? One method to solve this issue is called Frequency-Resolved Optical Gating, or FROG for short. This technique allows us to vary the path length of a split laser beam and recombine it with itself in a nonlinear medium. Once recombined you can observe the pulse on a spectrometer and use the software provided by the Trebino group to spectrally resolve the trace using their phase-retrieval algorithm. Doing this, we found that the main pulse has a full-width half maximum of about 89.700 fs with a phase of around 0.4 radians.

I. INTRODUCTION

To measure events in time, we need an event of a shorter time as a reference. This poses a difficulty when measuring the fastest event, or in other words, how do we measure the shortest event in time? Rick Trebino and his research group discovered a method to tackle this issue called Frequency-Resolved Optical Gating, or FROG[2] for short. The high-level idea is for a pulsed laser to combine with itself in a non-linear medium, in our case, a BBO crystal. This is done by passing the beam through a beam splitter, varying one of the beam's path lengths, and then recombining the two beams using a lens onto the crystal. The "optical gating" portion refers to the beams colliding in the crystal at the same time. To control this, one of the beams, known as the gate pulse, will travel a path where mirrors are mounted on a movable stage. This stage allows us to manipulate the travel distances until the two beams travel the same distance and then arrive at the crystal at the same time. The movable stage allows us to measure the spectrum of the signal at each delay. More generally, this is the "frequency resolved" portion. The other beam, the probe pulse, will only travel toward mirrors before arriving at the crystal. Both pulses are also focused onto the crystal using a lens so not only do the beams arrive at the crystal at the same time but also at the same point on the crystal.

Once the light arrives at the crystal at the same point and at the same time, the photons in the two beams interact such that a new photon emerges from the crystal with twice as much energy and half as much wavelength. In our experiment, we're using red laser light which collides with itself to give blue laser light. This blue light is what we're measuring on the spectrometer to give us the FROG Spectrogram or FROG trace. This trace is a function of frequency ω and delay τ where the trace is intensity. However, the input signal is most conveniently expressed in the time domain so we're measuring the Fourier Transform of the input signal. Specifically, we get the following expression as an equation for the FROG trace's intensity.

$$I_{FROG}(\omega,\tau) = \left| \int_{-\infty}^{\infty} E_{sig}(t,\tau) e^{-i\omega t}\, dt \right|^2 \tag{1}$$

The intensity equation is the squared absolute value of the Fourier Transform which means we've lost almost all information about the pulse's phase which is what we seek to retrieve using the Trebino group's phase-retrieval algorithm. The signal pulse, $E_{sig}(t,\tau)$, depends on the original pulse, $E(t)$, as well as this gate delay, $g(t-\tau)$ that we're implementing due to the moving stage. So, a better representation of the resultant intensity of the generated second harmonic due to the non-linear crystal is

$$I_{SHG\ FROG}(\omega,\tau) = \left| \int_{-\infty}^{\infty} E(t) g(t-\tau) e^{-i\omega t}\, dt \right|^2 \tag{2}$$

The main goal of our measurement is to fully characterize the pulse by determining the electric field of the pulse. From here, we collect the data from the spectrometer and process it to be used in the FROG "binner" and "frogger" programs provided by the Trebino group to visualize the FROG trace and analyze it. The "binner" program displays the trace and allows us to filter and crop the data such that it can be used by the "frogger" program. The "frogger" program is what performs the phase-retrieval algorithm to better characterize the pulse.

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II. METHODS

A. Experimental Setup

As mentioned before, we have built an optical setup similar to the one shown in FIG 1. We used a CARMEL CFL-04RFF 780nm Femtosecond laser and calibrated the stage's position such that the position of the stage, as it moves, is centred around where the two red beams recombine to produce a single blue beam. Once recombined, the beam passes through the crystal onto a white index card where the spectrometer probe is pointed so that the reflected light from the blue beam off of the card is measured. For our experiment, we moved the stage 0.8 millimetres in steps of 0.00125 millimetres resulting in the trace shown in FIG 3.

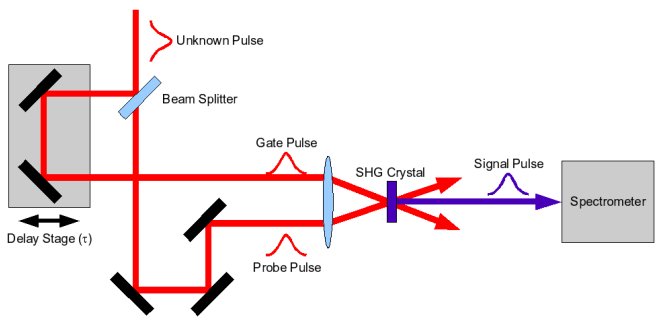


FIG. 1: Imaged here[1] is the general setup. The unknown laser pulse to observe is shot into a beam splitter where the two beams travel different distances. The gate pulse travels towards mirrors mounted on a movable stage before reflecting towards a lens. The movable stage allows us to systematically change the beam travel distance until the two beams travel equal distances. This is so the beams arrive at the crystal at the same time. The probe pulse only travels towards the lens. The two beams are focused through a lens so that not only do they arrive at the BBO crystal at the same time but also at the same point on the crystal. The output signal pulse is then analyzed with a spectrometer.

B. Data Processing

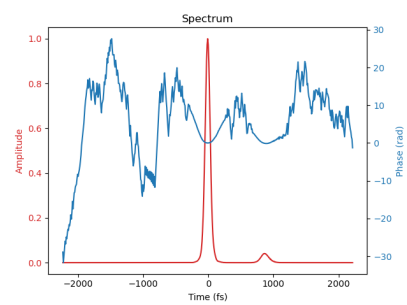
Before being able to use any of the data we've collected, we have to process it into a specific format required by the "binner" program. This is because we aren't using any Swamp Optics (the Trebino group's official company) peripherals for data collection. All this means is that we have to externally determine some values for the "binner" to recognize our data correctly. The data needs to be formatted with the size of the data, a temporal calibration in femtoseconds per pixel, a spectral calibration in nanometres per pixel, a central wavelength, and then the data itself. The temporal calibration is calculated using equation 3.

$$\tau = \frac{2 * d}{c} \tag{3}$$

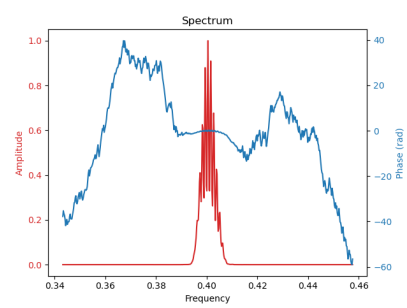
For every unit of distance that the stage has moved, you've doubled the laser's travel distance and you divide that by how fast it travels (the speed of light). The spectral calibration is calculated by taking the average of the sequential differences in wavelengths for the data. And lastly, the central wavelength is the wavelength that the "binner" will centre the trace around.

III. RESULTS

After using the "frogger" program, we are able to see our filtered and reconstructed trace found in FIG 3. Analyzing FIG 2a, we found that the secondary smaller pulse occurs within 1000 femtoseconds of the main pulse, however, due to the nature of the experiment it's unknown whether or not the secondary pulse is before or after the main pulse. We also can see the Fourier Transform of this estimated electric field in FIG 2b.



(a) Amplitude vs. Time



(b) Amplitude vs. Frequency

FIG. 2: Imaged here are the results of running the phase-retrieval algorithm where we have estimated the electric field's amplitude and phase in the time and frequency domain.

When measuring the full-width half-maximum of the amplitude signal in FIG 2a, we find the pulse duration

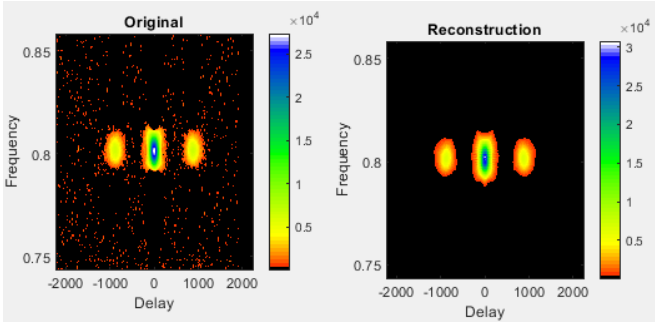


FIG. 3: Imaged here is the filtered trace and its reconstruction using the "frogger" program.

to be around 89.700 femtoseconds and a phase of around 0.4 radians.

Something that may have hindered our data is the fluorescence of the light as it reflected off the card. This fluorescence resulted in a large portion of our trace needing to be filtered. In doing so, a small portion of the actual pulse was filtered out. An optimal solution would be to mount the spectrometer so we're measuring the direct signal instead of a reflection which would remove possible fluorescence entirely.

IV. DISCUSSION

The FROG technique allows us to measure a pulse's intensity and estimate its phase. This is done by performing a spectrally resolved autocorrelation and the use of a phase-retrieval algorithm using the "frogger" program. Now that we have a functional FROG setup, the next step would be to pass the initial pulse through a piece of glass to further characterize the pulse. As hinted at before, the smaller secondary pulse is unknown if it occurs before or after the main one. Because different wavelengths of light diffract differently in glass, we would be able to determine whether the smaller secondary pulse that shows up in FIG 2a is actually before or after the main pulse. And when analyzing the pulse itself, we found it to have a full-width half maximum of 89.700 femtoseconds and a phase of around 0.4 radians.

[1] Frequency-resolved optical gating. URL `https://en.wikipedia.org/wiki/Frequency-resolved_optical_gating#/media/File:SHG_FROG.png`.

[2] S. O. LLC. Swamp optics tutorial frog. `https://www.swampoptics.com/assets/tutorials_frog.pdf`, 2015.

[3] R. Trebino, K. W. DeLong, D. N. Fittinghoff, J. N. Sweetser, M. A. Krumbügel, B. A. Richman, and D. J. Kane. Measuring ultrashort laser pulses in the time-frequency domain using frequency-resolved optical gating. *Rev. Sci. Instrum.*, 68(9):3277–3295, Sept. 1997.